

Intraday Momentum on SPY

A Mathematical Description of the Strategy and its Evaluation

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Based on Zarattini, Aziz & Barbon (2024), *Beat the Market*

This note states every step of the strategy as a formula, in the same order as the code in `src/intraday_momentum/`. Each section names the module that implements it.

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1 Notation and information structure

Days and minutes. Trading days are indexed by $t = 1, \dots, T$. The regular session (09:30–16:00 ET) is divided into $N = 390$ one-minute bars, indexed by $j = 0, \dots, N - 1$. Bar j covers the interval $[09:30 + j, 09:30 + j + 1)$. On half days the session ends after $N_t < N$ bars; in general $N_t \leq N$ is the number of bars on day t (`data.py`).

Prices. For each day and bar we observe open, high, low, close and volume $(O_{t,j}, H_{t,j}, L_{t,j}, C_{t,j}, V_{t,j})$. We write

$$P_{t,k} := C_{t,k-1}, \quad k = 1, \dots, N_t,$$

for the price observed at minute k after the open (the close of the bar that ends at $09:30 + k$). The day's open and close are

$$O_t := O_{t,0}, \quad C_t := P_{t,N_t}.$$

Filtration. Let $\mathcal{F}_{t,k}$ be the information available at minute k of day t : all bars of days $1, \dots, t-1$ and bars $0, \dots, k-1$ of day t . A backtest is free of look-ahead bias if and only if every decision taken at (t, k) is $\mathcal{F}_{t,k}$ -measurable and executed at a price that is realised *after* (t, k) . Every quantity below is marked with the information set it depends on.

2 The Noise Area (signals.py)

2.1 Average move from the open

On every past day $t-i$ we measure the absolute relative move from the open after k minutes:

$$m_{t-i,k} = \left| \frac{P_{t-i,k}}{O_{t-i}} - 1 \right|. \quad (1)$$

The *expected size of a normal move* at minute k is estimated by the mean over the previous $n = 14$ days (the lookback):

$$\sigma_{t,k} = \frac{1}{n} \sum_{i=1}^n m_{t-i,k}. \quad (2)$$

Since the sum starts at $i = 1$, $\sigma_{t,k}$ only uses days before t , so it is $\mathcal{F}_{t,0}$ -measurable. In the code this is a rolling mean over days followed by a shift of one day.

Remark (Why the band widens like $\sqrt{k}$). $\sigma_{t,k}$ is a mean absolute move, not a standard deviation. Suppose one-minute log returns are i.i.d. $\mathcal{N}(0, s^2)$. Then $\ln(P_{t,k}/O_t) \sim \mathcal{N}(0, ks^2)$, and for small moves

$$\mathbb{E}[m_{t,k}] \approx \mathbb{E}|Z| s\sqrt{k} = \sqrt{\frac{2}{\pi}} s\sqrt{k}, \quad Z \sim \mathcal{N}(0, 1).$$

So the threshold for a “significant” move grows like $\sqrt{k}$. This gives the concave, funnel-shaped Noise Area in Figure 1 of the paper. A fixed threshold would be too loose in the morning and too tight in the afternoon.

2.2 Boundaries with gap adjustment

Let C_{t-1} be the previous close and D_t the cash dividend if t is an ex-dividend date ($D_t = 0$ otherwise). The dividend-adjusted previous close is $\tilde{C}_{t-1} = C_{t-1} - D_t$. With the volatility multiplier VM (paper: VM = 1) the boundaries are

$$\text{UB}_{t,k} = \max(O_t, \tilde{C}_{t-1}) (1 + \text{VM} \sigma_{t,k}), \quad (3)$$

$$\text{LB}_{t,k} = \min(O_t, \tilde{C}_{t-1}) (1 - \text{VM} \sigma_{t,k}). \quad (4)$$

The Noise Area is the interval $\mathcal{N}_{t,k} = [\text{LB}_{t,k}, \text{UB}_{t,k}]$.

Remark (Gap adjustment). After a gap down ($O_t < \tilde{C}_{t-1}$) the upper band is anchored at yesterday’s close. A long signal then requires the price to close the whole gap and move an extra $\sigma_{t,k}$. The lower band stays anchored at the open, so continuation of the gap is detected quickly. Gap ups work the same way in the other direction. Without the dividend correction, the mechanical price drop on ex-dates would look like a gap down.

2.3 Session VWAP

With the typical price $\bar{p}_{t,j} = (H_{t,j} + L_{t,j} + C_{t,j})/3$,

$$\text{VWAP}_{t,k} = \frac{\sum_{j=0}^{k-1} \bar{p}_{t,j} V_{t,j}}{\sum_{j=0}^{k-1} V_{t,j}}, \quad (5)$$

which uses bars up to minute k only, so it is $\mathcal{F}_{t,k}$ -measurable.

3 Signal and position (backtest.py)

3.1 Decision times

Positions can only change at the decision times

$$\mathcal{K}_t = \{30, 60, \dots\} \cap \{1, \dots, N_t - 1\},$$

i.e. 10:00, 10:30, ..., 15:30 on a full day. Let $x_{t,k} \in \{-1, 0, +1\}$ be the position (short, flat, long) held after the decision at minute k . Each day starts flat, $x_{t,0} = 0$, and between decisions the position is unchanged.

3.2 Base model: stop at the opposite band

At $k \in \mathcal{K}_t$, with x^- = the current position:

$$x_{t,k} = \begin{cases} +1 & \text{if } P_{t,k} > \text{UB}_{t,k}, \\ -1 & \text{if } P_{t,k} < \text{LB}_{t,k}, \\ x^- & \text{otherwise.} \end{cases} \quad (6)$$

This rule has *memory*: inside the Noise Area the previous position is kept. A long is only closed (and immediately reversed) when the price crosses the *lower* band.

3.3 Extension 1: trailing stop at the current band and VWAP

$$x_{t,k} = \begin{cases} +1 & \text{if } P_{t,k} > \max(\text{UB}_{t,k}, \text{VWAP}_{t,k}), \\ -1 & \text{if } P_{t,k} < \min(\text{LB}_{t,k}, \text{VWAP}_{t,k}), \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

This rule is *memoryless*. As soon as a long falls back below the upper band or below VWAP, it is closed. This is exactly a trailing stop at $\max(\text{UB}_{t,k}, \text{VWAP}_{t,k})$.

3.4 Execution

A decision at (t, k) is executed at the open of the next bar,

$$\hat{P}_{t,k} = O_{t,k} \quad (\text{realised after } \mathcal{F}_{t,k}),$$

and every open position is closed at the day's close C_t . In particular $x_{t,N_t} = 0$: no overnight exposure.

4 Position sizing

The number of shares is fixed at the open of each day:

$$q_t = \left\lfloor \frac{A_{t-1} L_t}{O_t} \right\rfloor, \quad (8)$$

where A_{t-1} is the equity at the end of day $t - 1$ and L_t is the leverage.

Full exposure (base, Extension 1). $L_t = 1$.

Volatility targeting (Extension 2). Let $r_s = C_s/C_{s-1} - 1$ be daily close-to-close returns and

$$\hat{\sigma}_t = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (r_{t-i} - \bar{r}_t)^2}, \quad \bar{r}_t = \frac{1}{n} \sum_{i=1}^n r_{t-i}, \quad (9)$$

the realised daily volatility over the previous $n = 14$ days ($\mathcal{F}_{t,0}$ -measurable). Then

$$L_t = \min\left(L_{\max}, \frac{\sigma^*}{\hat{\sigma}_t}\right), \quad \sigma^* = 2\%, \quad L_{\max} = 4. \quad (10)$$

Exposure is reduced when the market is volatile and increased when it is calm. Since SPY's daily volatility is typically around 1%, the typical leverage is about 2.

5 Profit and loss with transaction costs

Let the day's round trips be $\ell = 1, \dots, R_t$, each with side $s_\ell \in \{\pm 1\}$, entry price e_ℓ and exit price z_ℓ . Every entry, exit or reversal leg is one order of q_t shares. Let M_t be the number of orders. A reversal from $+1$ to -1 counts as two orders.

Gross and net P&L.

$$\Pi_t^{\text{gross}} = \sum_{\ell=1}^{R_t} s_\ell q_t (z_\ell - e_\ell), \quad \Pi_t = \Pi_t^{\text{gross}} - M_t \kappa(q_t), \quad (11)$$

with the cost of one order

$$\kappa(q) = \max(c_{\min}, cq) + \delta q, \quad (12)$$

where $c = \$0.0035$ is the commission per share, $c_{\min} = \$0.35$ the minimum commission per order and δ the slippage per share ($\$0.001$ in the paper scenario, $\$0.005$ in the conservative one).

Equity and returns.

$$A_t = A_{t-1} + \Pi_t, \quad R_t^{\text{strat}} = \frac{\Pi_t}{A_{t-1}}, \quad A_0 = \$100,000. \quad (13)$$

Proposition (Break-even cost). *Let $\bar{\pi}$ be the average gross profit per share per round trip. Each round trip has two orders. For large q (where the minimum commission does not bind), the strategy is profitable on average iff*

$$\bar{\pi} > 2(c + \delta).$$

Proof. A round trip earns $q\bar{\pi}$ on average and costs $2\kappa(q) = 2q(c + \delta)$ when $cq \geq c_{\min}$. $\square$

The paper reports $\bar{\pi} \approx \$0.09$, so the edge vanishes at roughly $c + \delta \approx \$0.045$ per share and side. This is the reason the cost assumptions matter so much for an intraday strategy.

6 Performance metrics (metrics.py)

Let $R_1, \dots, R_T$ be daily returns, with $\bar{R}$ their mean and s_R their sample standard deviation. We use 252 trading days per year.

$$\text{Total return} = \prod_{t=1}^T (1 + R_t) - 1, \quad (14)$$

$$\text{Annualised return (CAGR)} = \left(\prod_{t=1}^T (1 + R_t) \right)^{252/T} - 1, \quad (15)$$

$$\text{Annualised volatility} = s_R \sqrt{252}, \quad (16)$$

$$\text{Sharpe ratio} = \frac{\bar{R} - r_f/252}{s_R} \sqrt{252} \quad (r_f = 0 \text{ in our tables}), \quad (17)$$

$$\text{Max drawdown} = \max_t \left(1 - \frac{E_t}{\max_{u \leq t} E_u} \right), \quad E_t = \prod_{u \leq t} (1 + R_u), \quad (18)$$

$$\text{Hit ratio} = \frac{\#\{t : R_t > 0\}}{\#\{t : R_t \neq 0\}}. \quad (19)$$

Remark (Why $\sqrt{252}$). If daily returns are independent with variance s^2 , the variance of a sum over 252 days is $252 s^2$. The standard deviation therefore scales with $\sqrt{252}$, while the mean scales with 252. Hence the Sharpe ratio scales with $252/\sqrt{252} = \sqrt{252}$.

Proposition (Leverage does not change the Sharpe ratio). For $r_f = 0$ and any constant $\lambda > 0$: $\text{SR}(\lambda R) = \text{SR}(R)$.

Proof. $\overline{\lambda R} = \lambda \bar{R}$ and $s_{\lambda R} = \lambda s_R$, so the ratio is unchanged. $\square$

Therefore strategies should be compared by their Sharpe ratio and not by total return. Volatility targeting changes L_t over time, so it *can* change the Sharpe ratio. Most of the increase in total return from Extension 2, however, comes from the average leverage.

Alpha and beta. OLS regression of strategy returns on SPY returns,

$$R_t^{\text{strat}} = \alpha + \beta R_t^{\text{SPY}} + \varepsilon_t, \quad \hat{\theta} = (X^\top X)^{-1} X^\top y, \quad (20)$$

with $X = [\mathbf{1}, R^{\text{SPY}}]$. Standard errors are $\widehat{\text{se}}(\hat{\theta}_i) = \sqrt{\hat{s}_\varepsilon^2 [(X^\top X)^{-1}]_{ii}}$ with $\hat{s}_\varepsilon^2 = \hat{\varepsilon}^\top \hat{\varepsilon} / (T - 2)$, and $t_i = \hat{\theta}_i / \widehat{\text{se}}(\hat{\theta}_i)$. We report the annualised alpha $252 \hat{\alpha}$.

Statistical significance of the Sharpe ratio. The t -statistic of the mean daily return is $t = \bar{R} / (s_R / \sqrt{T}) = \text{SR}_{\text{daily}} \sqrt{T}$. With $T = 252Y$ days (Y years) this gives

$$t \approx \text{SR}_{\text{ann}} \sqrt{Y}. \quad (21)$$

For example, a Sharpe ratio of 1 over 4 test years gives $t \approx 2$, which is only borderline significant.

7 Evaluation protocol

- **Split.** The sample is divided chronologically into a training period $\mathcal{T}_{\text{train}}$ (2016–2021) and a test period $\mathcal{T}_{\text{test}}$ (2022–today). Each period is simulated separately with $A_0 = \$100,000$.
- **No fitting on the test set.** All variants use the paper’s parameters $(n, \text{VM}, \sigma^*, L_{\text{max}}) = (14, 1, 2\%, 4)$. The grid $n \in \{7, 14, 30, 60\}$, $\text{VM} \in \{0.8, 1.0, 1.2, 1.5\}$ is evaluated on $\mathcal{T}_{\text{train}}$ only, as a robustness check. A good result is a flat plateau of Sharpe ratios rather than one sharp peak.
- **No look-ahead.** $\sigma_{t,k}$, $\hat{\sigma}_t$, $\tilde{C}_{t-1}$ and L_t are $\mathcal{F}_{t,0}$ -measurable. $P_{t,k}$ and $\text{VWAP}_{t,k}$ are $\mathcal{F}_{t,k}$ -measurable. Execution happens at $O_{t,k}$, which lies after $\mathcal{F}_{t,k}$. These properties are checked by unit tests (`tests/test_signals.py`).
- **Costs.** Two scenarios ($\delta = 0.001$ and $\delta = 0.005$), plus a sweep over $c + \delta \in [0.0035, 0.0235]$ in the test period.

8 Summary of parameters

Symbol	Meaning	Value
n	lookback for $\sigma_{t,k}$ and $\hat{\sigma}_t$	14 days
VM	volatility multiplier	1
$\mathcal{K}_t$	decision times	every 30 min from 10:00
σ^*	daily volatility target	2%
$L_{\max}$	leverage cap	4
c	commission per share	\$0.0035
$c_{\min}$	minimum commission per order	\$0.35
δ	slippage per share	\$0.001 / \$0.005
A_0	initial capital	\$100,000